Two springs in series, one of them damageable: analytical, static and dynamic solutions

Repository `1D_damage_spring_system`

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1 Problem statement

We consider two springs (elastic bars) in series:

$$\text{node 0 (clamped)} \xrightarrow{\text{spring 1, } k_1^d(u_1)} \text{node 1 (mass } m_1) \xrightarrow{\text{spring 2, } k_2} \text{node 2 } (u_2 \text{ imposed})$$

Spring 1 is damageable, spring 2 remains linear elastic. The loading is displacement controlled: u_2 is imposed at node 2, increasing from 0 to u_2^f. The unknown is the displacement u_1 of the intermediate node. Spring 1 softens with damage, which produces a *snap-back*: the equilibrium branch $F(u_2)$ turns back in u_2, and quasi-static displacement control cannot follow it. We compare:

1. the **analytical** solution (closed form, complete branch including the unstable part);
2. the **static** solution (Newton–Raphson at each u_2 increment);
3. the **dynamic** solution (mass m_1 at node 1, implicit Newmark scheme, Newton–Raphson at each time step).

The focus of the comparison is the shape of the residual $R(u_1)$ that the Newton–Raphson has to cancel, and the way the inertia term regularises its non-monotonicity.

1.1 Damage law

The damage of spring 1 is a variable $d \in [0, 1]$ depending on the maximum displacement reached $\kappa = \max_{\tau \le t} u_1(\tau)$ (history variable, irreversibility):

$$d(\kappa) = 1 - \exp\left(-\frac{\kappa^2}{u_c^2}\right), \qquad k_1^d(u_1) = k_1\,(1 - d(\kappa)), \qquad F_1 = k_1^d\,u_1. \tag{1}$$

On loading ($u_1 = \kappa$), $d'(u_1) = \dfrac{2u_1}{u_c^2}\exp(-u_1^2/u_c^2)$; on unloading ($u_1 < \kappa$), d is frozen and the spring unloads elastically with the secant stiffness $k_1(1-d)$. For monotonic loading (analytical and static cases), $\kappa = u_1$ at all times.

1.2 Parameters

Consistent unit system N / mm / s, hence mass in tonne (1 t = 1000 kg).

Spring 1	$E_1 = 10\,000$ MPa, $A_1 = 20$ mm^2, $L_1 = 5$ mm $u_c = 5$ mm	$k_1 = E_1A_1/L_1 = 40\,000$ N/mm
Spring 2	$E_2 = 20\,000$ MPa, $A_2 = 30$ mm^2, $L_2 = 500$ mm	$k_2 = E_2A_2/L_2 = 1\,200$ N/mm
Mass	$m_1 = 10$ t (node 1)	
Loading	$u_2 : 0 \to u_2^f = 80$ mm	
Newton–Raphson	tolerance $\lvert R\rvert \le 10^{-6}$ N, 50 iterations max	

The ratio $k_1/k_2 = 33.3$: spring 1 is very stiff while intact, and almost all of the imposed displacement is absorbed by spring 2 ($u_1 \simeq u_2\, k_2/(k_1+k_2)$). The mass $m_1 = 10$ t is not physical for a $20\,\mathrm{mm}^2 \times 5$ mm bar (actual mass $\rho A_1 L_1 \sim 10^{-6}$ t): it is a numerical regularisation parameter, see §5.

2 From the mass–spring system to the residuals

2.1 Equation of motion of node 1

The forces acting on the mass m_1 are the force of spring 1, $f_1 = k_1(1-d)\,u_1$, directed towards node 0, and the force of spring 2, $f_2 = k_2(u_2 - u_1)$, directed towards node 2. Newton's second law gives

$$m_1\ddot{u}_1 = -k_1(1-d)\,u_1 + k_2(u_2(t) - u_1) \quad \Longleftrightarrow \quad m_1\ddot{u}_1 + \underbrace{(k_1^d(u_1) + k_2)u_1}_{f^{\mathrm{int}}(u_1)} - \underbrace{k_2\,u_2(t)}_{f^{\mathrm{ext}}(t)} = 0. \tag{2}$$

In the notation of the companion note on nonlinear dynamics, $M = m_1$, $C = 0$, $f^{\mathrm{int}}(u_1) = (k_1^d + k_2)u_1$ and $f^{\mathrm{ext}}(t) = k_2 u_2(t)$: the imposed displacement enters as an external force. This is the usual finite element treatment of a prescribed degree of freedom. Writing the two-node system

$$\begin{bmatrix} k_1^d + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ F_2 \end{Bmatrix}, \tag{3}$$

the first row is the equilibrium of node 1 with the known term $k_2 u_2$ moved to the right-hand side, and the second row gives the reaction $F_2 = k_2(u_2 - u_1)$, computed after the solve.

2.2 Damage law and its derivative

With $x = u_1/u_c$ and $\kappa = u_1$ on loading,

$$k_1^d(u_1) = k_1 e^{-x^2}, \qquad \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} = -\frac{2k_1 u_1}{u_c^2} e^{-x^2}, \qquad \frac{\mathrm{d}F_1}{\mathrm{d}u_1} = k_1^d + u_1 \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} = k_1 e^{-x^2}(1 - 2x^2). \tag{4}$$

On unloading ($u_1 < \kappa$), k_1^d is constant and its derivative is zero.

2.3 Dynamic residual (Newmark)

Substituting the Newmark relation $\ddot{u}_1^{n+1} = c_1(u_1^{n+1} - \tilde{u})$, $c_1 = 1/(\beta \Delta t^2)$, into (2) at time t^{n+1} gives the residual and tangent of the dynamic problem:

$$r_{\mathrm{dyn}}(u_1) = m_1 c_1\,(u_1 - \tilde{u}) + (k_1^d(u_1) + k_2)u_1 - k_2 u_2^{n+1}, \qquad \frac{\partial r_{\mathrm{dyn}}}{\partial u_1} = m_1 c_1 + k_t(u_1), \tag{5}$$

with k_t the static tangent defined below.

2.4 Static residual and Newton–Raphson

The quasi-static case is the limit $m_1\ddot{u}_1 \to 0$ (equivalently $m_1 = 0$ in (5)):

$$R(u_1; u_2) = (k_1^d(u_1) + k_2)u_1 - k_2 u_2 = 0, \tag{6}$$

$$k_t(u_1) = \frac{\partial R}{\partial u_1} = k_1^d + u_1 \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} + k_2 = k_1 e^{-x^2}(1 - 2x^2) + k_2 \quad \text{(loading)}. \tag{7}$$

Newton–Raphson: $u_1^{(i+1)} = u_1^{(i)} - R(u_1^{(i)})/k_t(u_1^{(i)})$, starting from the solution of the previous increment. The dynamic residual (5) is simply R augmented by $m_1 c_1(u_1 - \tilde{u})$.

2.5 Snap-back condition and multiplicity of roots

R is non-monotonic in u_1 as soon as k_t changes sign. The steepest negative slope of spring 1 is obtained from $\mathrm{d}^2F_1/\mathrm{d}u_1^2 = 0$, i.e. $x^2 = 3/2$, where $\mathrm{d}F_1/\mathrm{d}u_1 = -2e^{-3/2}k_1$. Hence

$$\text{snap-back} \iff k_2 < 2e^{-3/2}k_1 \simeq 0.446\,k_1. \tag{8}$$

In that case $k_t < 0$ on an interval $[u_1^{\lim}, u_1^{\lim\prime}]$, the residual has up to three roots for a given u_2, and the Newton–Raphson started just before the local maximum of R cycles. Inertia adds m_1c_1 to the tangent: if $m_1c_1 > -\min k_t$, r_{dyn} is strictly increasing and has a single root.

2.6 Closed-form branch

R is linear in u_2, so the equilibrium branch is obtained by parametrising with u_1 (monotonic loading, $\kappa = u_1$):

$$u_2(u_1) = u_1\Big(1 + \frac{k_1}{k_2}e^{-x^2}\Big), \qquad F(u_1) = k_2(u_2 - u_1) = k_1u_1e^{-x^2}. \tag{9}$$

- **Force peak**: $\mathrm{d}F/\mathrm{d}u_1 = 0 \Rightarrow u_1^{\mathrm{peak}} = u_c/\sqrt{2}$, $F_{\max} = k_1u_c/\sqrt{2e}$, $d^{\mathrm{peak}} = 1 - e^{-1/2} = 0.393$. This is what sets u_c from the tensile strength: $u_c = \sqrt{2e}\,f_tA_1/k_1$.
- **Limit point** (turning point of u_2): $\mathrm{d}u_2/\mathrm{d}u_1 = k_t/k_2 = 0$, root of $e^{-x^2}(2x^2 - 1) = k_2/k_1$ on $x \in [1/\sqrt{2}, \sqrt{3/2}]$. Displacement control cannot go beyond $u_2^{\lim}$; it jumps at fixed u_2 to the root on the damaged branch.
- **Damaged branch**: for $x \gg 1$, $k_1^d \to 0$, $u_2 \to u_1$, $F \to 0$.
- **Energies**: free energy $\Psi = \frac{1}{2}k_1(1-d)u_1^2 + \frac{1}{2}k_2(u_2-u_1)^2$, energy release rate $Y = \frac{1}{2}k_1u_1^2$, dissipation $\mathcal{D} = \int Y\,\mathrm{d}d$. Along the branch $\int F\,\mathrm{d}u_2 = \Psi_2 - \Psi_1 + \mathcal{D}_{\mathrm{branch}}$; at the jump ($u_2$ constant, zero work) the released energy $\Psi_1 - \Psi_2 = \mathcal{D}_{\mathrm{branch}} + |W|$ becomes kinetic energy. The total fracture energy is $\int_0^\infty F\,\mathrm{d}u_1 = \frac{1}{2}k_1u_c^2$, i.e. $G_f = k_1u_c^2/2A_1$.
- **Free vibration after rupture**: $m_1\ddot{u}_1 + k_2(u_1 - u_2^f) = 0$, period $2\pi\sqrt{m_1/k_2}$, amplitude $u_2^{\lim} - u_1^{\lim}$ without damping.

3 Analytical solution

The equilibrium of node 1 reads

$$R(u_1; u_2) = k_1^d(u_1)\,u_1 - k_2\,(u_2 - u_1) = u_1(k_1^d(u_1) + k_2) - k_2u_2 = 0. \tag{10}$$

Rather than solving (10) for u_1 at given u_2 (a transcendental equation with several roots), we parametrise the equilibrium branch by u_1:

$$u_2(u_1) = u_1\left(1 + \frac{k_1^d(u_1)}{k_2}\right), \qquad F(u_1) = k_2\,(u_2 - u_1) = k_1^d(u_1)\,u_1 = k_1\,u_1\,e^{-u_1^2/u_c^2}. \tag{11}$$

With $x = u_1/u_c$, the tangent stiffness of spring 1 is $\mathrm{d}F/\mathrm{d}u_1 = k_1e^{-x^2}(1 - 2x^2)$ and the system tangent is $k_t(u_1) = k_1e^{-x^2}(1 - 2x^2) + k_2$. Hence:

- **force peak** at $x = 1/\sqrt{2}$: $u_1 = u_c/\sqrt{2} = 3.536$ mm, $F_{\max} = k_1u_ce^{-1/2}/\sqrt{2} = 85\,776$ N, $u_2 = 75.02$ mm;
- **limit point** (snap-back) where $\mathrm{d}u_2/\mathrm{d}u_1 = k_t/k_2 = 0$: $u_1 = 3.624$ mm, $u_2^{\lim} = 75.06$ mm; beyond it, u_2 decreases along the equilibrium branch down to $u_2 \simeq 13.3$ mm and then rises again on the fully damaged branch ($d \to 1$, $F \to 0$, $u_2 \simeq u_1$).

Quasi-static displacement control therefore cannot go beyond $u_2^{\lim}$ without jumping, at fixed u_2, from the state ($u_1 = 3.62,\ F = F_{\max}$) to the state ($u_1 \simeq u_2,\ F \simeq 0$).

Energy balance of the jump. With the free energy $\Psi = \frac{1}{2}k_1(1-d)u_1^2 + \frac{1}{2}k_2(u_2-u_1)^2$ and the damage dissipation $\mathcal{D} = \int Y\,\mathrm{d}d$, $Y = \frac{1}{2}k_1 u_1^2$, the work of the force along the equilibrium branch between the two states of equal u_2 satisfies $W = \int F\,\mathrm{d}u_2 = \Psi_2 - \Psi_1 + \mathcal{D}_{\text{branch}}$. Since the jump occurs at constant u_2 ($W = 0$), the released energy is $\Psi_1 - \Psi_2 = \mathcal{D}_{\text{branch}} + |W|$. Numerically: $\Psi_1 - \Psi_2 = 3\,217$ J, $\mathcal{D}_{\text{branch}} = 455$ J, $|W| = 2\,762$ J. The area $|W|$ between the vertical jump and the equilibrium branch is thus the energy released as kinetic energy at the jump, in excess of what the quasi-static branch would have dissipated by damage. (The script calls it `released_energy`.)

4 Static solution

u_2 is imposed over 2000 uniform increments and (10) is solved by Newton–Raphson with the analytical tangent

$$k_t(u_1) = k_1^d(u_1) + u_1\,\frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} + k_2, \qquad u_1^{(i+1)} = u_1^{(i)} - \frac{R(u_1^{(i)})}{k_t(u_1^{(i)})}, \tag{12}$$

starting from the solution of the previous increment. The residual (10) is not monotonic in u_1 ($k_t < 0$ for $u_1 \in [3.62\,;\simeq 12]$ mm, minimum $k_t = -16\,650$ N/mm): at a given u_2 it may have one or three roots (figure 2, left).

Behaviour at the limit point. For $u_2 > u_2^{\lim}$ there is no longer a root in the neighbourhood of the previous iterate: the Newton–Raphson cycles around the local maximum of the residual without converging, or, with coarse increments, may take a huge step from the flat top of the residual and land by accident on the damaged branch. The optional backtracking line search (`parameters.line_search`) removes this erratic behaviour: the Newton direction is kept but the step length $\alpha \in (0, 1]$ is halved until $|R|$ decreases, so the iterates climb to the local maximum $R(u_1^{\lim}; u_2) = -k_2(u_2 - u_2^{\lim})$ and stall there in a few iterations, whatever the increment size. The solver checks convergence and, on failure (stall or `max_iter`), stops the quasi-static calculation at that increment with a message: no restart on the damaged branch and no arc-length continuation are used, so the static curve ends at the limit point (figure 3). This is the genuine behaviour of displacement control without any numerical device; the dynamic solution of §5 is the physical continuation. Away from the limit point, a few iterations suffice.

5 Dynamic solution

A mass m_1 is added at node 1; node 2 remains displacement controlled by a ramp $u_2(t) = v_2\,t$ up to u_2^f, then held at u_2^f for n_{hold} steps to let the jump complete. The equation of motion of node 1 (undamped) is

$$m_1\ddot{u}_1 + k_1^d(u_1)\,u_1 - k_2\,(u_2(t) - u_1) = 0. \tag{13}$$

Newmark scheme. Implicit Newmark, $\beta = \frac{1}{4}$, $\gamma = \frac{1}{2}$ (average acceleration, unconditionally stable, no numerical damping), in predictor–corrector form:

$$\tilde{u} = u_n + \Delta t\,v_n + \tfrac{\Delta t^2}{2}(1-2\beta)\,a_n, \qquad \tilde{v} = v_n + (1-\gamma)\Delta t\,a_n, \tag{14}$$

$$a_{n+1} = c_1\,(u_{n+1} - \tilde{u}), \quad c_1 = \frac{1}{\beta\Delta t^2}, \qquad v_{n+1} = \tilde{v} + \gamma\Delta t\,a_{n+1}. \tag{15}$$

For a general damped system $M\ddot{u} + C\dot{u} + f^{\text{int}}(u) - f^{\text{ext}}(t) = 0$, substituting both Newmark relations gives the residual and its tangent

$$r(u^{n+1}) = \Big(\frac{M}{\beta\Delta t^2} + \frac{C\gamma}{\beta\Delta t}\Big)(u^{n+1} - \tilde{u}) + C\tilde{v} + f^{\text{int}}(u^{n+1}) - f^{\text{ext}}(t^{n+1}), \tag{16}$$

$$\frac{\partial r}{\partial u} = \frac{M}{\beta\Delta t^2} + \frac{C\gamma}{\beta\Delta t} + \frac{\partial f^{\text{int}}}{\partial u}. \tag{17}$$

Here $C = 0$, so the residual to cancel at step $n+1$ and its tangent are those of the static case augmented by the inertia term:

$$R_{\text{dyn}}(u_1) = m_1 c_1\,(u_1 - \tilde{u}) + R(u_1; u_2^{n+1}), \qquad k_t^{\text{dyn}}(u_1) = m_1 c_1 + k_t(u_1). \tag{18}$$

The Newton–Raphson is initialised with the predictor $\tilde{u}$. After convergence, a_{n+1} and v_{n+1} are updated with the corrector relations; at $t = 0$, $a_0 = (f^{\text{ext}}(0) - f^{\text{int}}(u_0))/m_1$. The history variable κ is frozen during the iterations of a step and updated after convergence.

Choice of the time step: inertial regularisation. From (18), inertia adds to the tangent the constant $m_1 c_1 = m_1/(\beta\Delta t^2)$. If $m_1 c_1 > -\min k_t = 16\,650$ N/mm, the dynamic residual is strictly increasing in u_1: a single root, unambiguous Newton–Raphson, no more abrupt jump at the limit point. With the natural pulsation of the intact system $\omega_0 = \sqrt{(k_1 + k_2)/m_1} = 64.2$ rad/s ($T_0 = 97.9$ ms) and $\Delta t = T_0/N$, the inertia/stiffness ratio is

$$\frac{m_1 c_1}{k_1 + k_2} = \frac{N^2}{4\pi^2\beta} = \frac{N^2}{\pi^2} \quad (\beta = \tfrac{1}{4}). \tag{19}$$

A fine step ($N = 40$, initial choice) gives a ratio $\simeq 160$: the dynamic residual is an almost perfect straight line and the comparison with statics loses its interest. We keep $N = 5$, i.e. $\Delta t = 19.6$ ms, $m_1 c_1 = 104\,360$ N/mm, ratio 2.5: enough to regularise (minimum dynamic tangent +87 700 N/mm) without flattening the contribution of the damaged spring. The ramp lasts 150 steps (2.94 s, i.e. 30 natural periods: loading slow compared with the system dynamics), followed by 100 hold steps. The Newton–Raphson converges in at most 3 iterations.

Results. The dynamic solution follows the analytical equilibrium branch beyond the limit point (u_2 up to $\simeq 78$ mm before F drops, figure 1), then node 1 "jumps" to the damaged branch during the hold. Since the released energy (§3) has no dissipation mechanism (no damping, spring 1 already fully damaged), the mass then oscillates freely on spring 2 about $u_1 = u_2^f$ with period $2\pi\sqrt{m_1/k_2} = 0.574$ s and amplitude $\simeq 73$ mm (figure 4); F_2 then becomes alternately negative (spring 2 in compression). Thanks to irreversibility (κ), damage stays at $d = 1$ during these oscillations; with the reversible law $d = d(u_1)$ of the original scripts, d dropped back to 0.66 at the low point ($u_1 \simeq 5$ mm) and spring 1 "healed", which is non-physical.

6 Figures

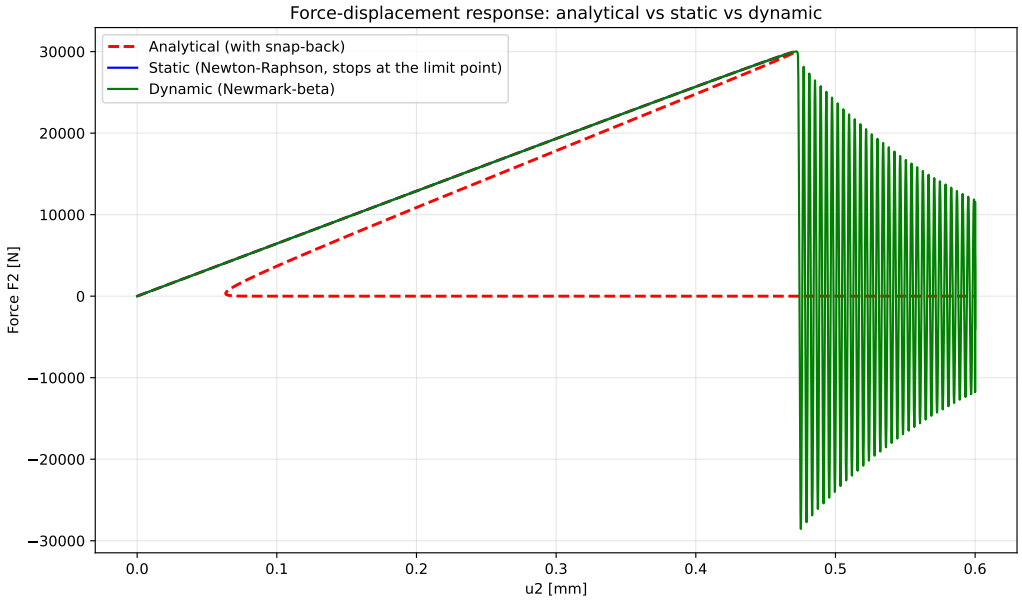


Figure 1: Force–displacement response $F_2(u_2)$: analytical (complete branch with snap-back), static (stops at the limit point) and dynamic (passes the limit point, then jumps and oscillates).

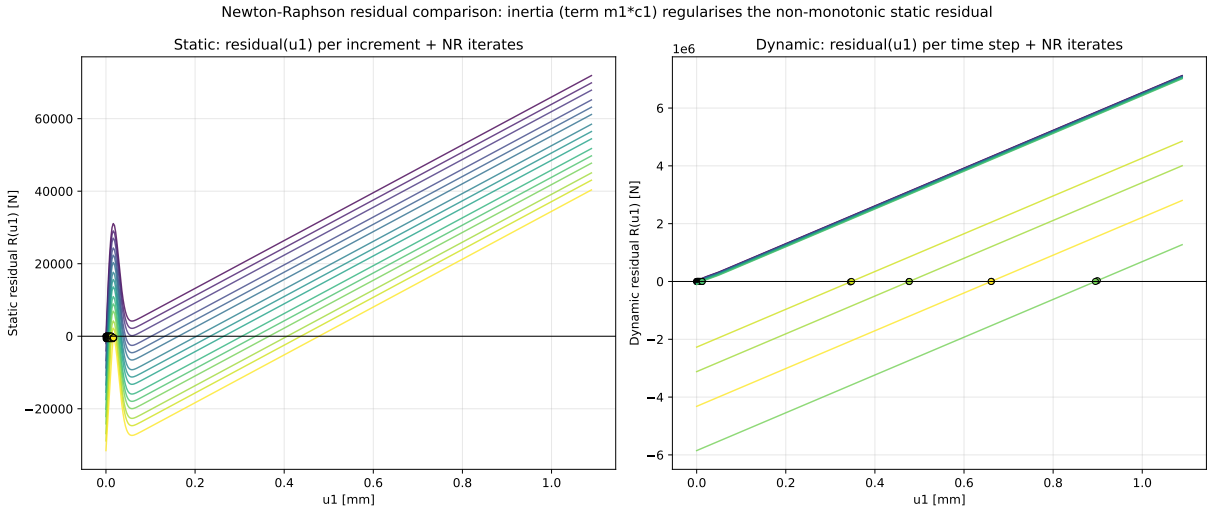


Figure 2: Residual $R(u_1)$ to be cancelled by the Newton–Raphson, one curve per increment (colour from purple, start of loading, to yellow, end) with the iterates as dots. Left: static, non-monotonic residual (bump due to the softening of spring 1), up to three roots. Right: dynamic, the term $m_1 c_1(u_1 - \tilde{u})$ makes the residual strictly increasing; the curves shift vertically with $\tilde{u}$, which carries the kinematic history.

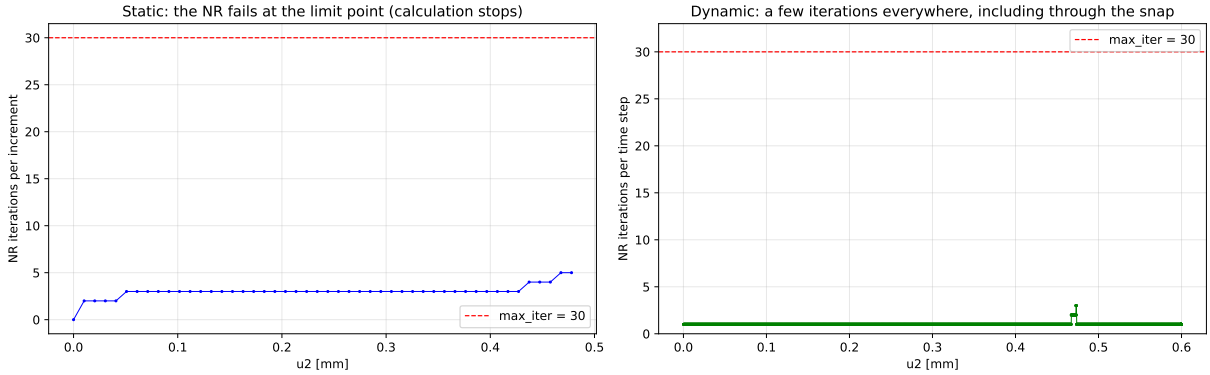


Figure 3: Number of Newton–Raphson iterations per increment (static, left) and per time step (dynamic, right). In statics the count explodes to `max_iter` at the limit point and the calculation stops; in dynamics a few iterations suffice everywhere, including through the snap. This is the practical justification of the dynamic approach.

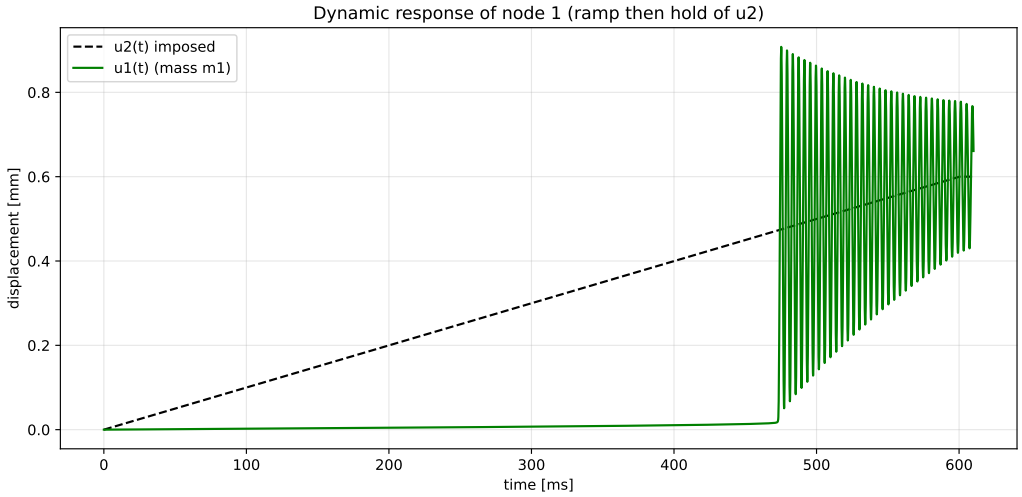


Figure 4: Dynamic time response: ramp of u_2 then hold; jump of node 1 and undamped free oscillation on spring 2.

7 Script layout and checks

`parameters.py`	physical and numerical parameters, derived quantities
`model.py`	damage law with κ, static residual/tangent, Newton–Raphson
`analytical_solution.py`	branch (11), peak, energy balance
`static_solution.py`	u_2 increments, Newton–Raphson, stop at the limit point, iterate history
`dynamic_solution.py`	predictor–corrector Newmark, ramp + hold, iterate history
`comparison.py`	assembles the three solutions, figures (`-save` writes them to `figures/`)

Checks performed: numerical peak and limit point consistent with the analytical values; identity $\Psi_1 - \Psi_2 = \mathcal{D}_{\text{branch}} + |W|$ satisfied; Newton–Raphson convergence at each increment/step (tolerance 10^{-6} N; a tolerance of 10^{-9} N was below the round-off noise of the inertia terms $\sim 10^7$ N); positivity of the dynamic tangent; period of the free oscillations; conservation of $d = 1$ after the jump.

Limitations and extensions. Non-physical mass (regularisation parameter); no damping, hence perpetual oscillations after the jump (a damping $c\,\dot{u}_1$ or a dissipative Newmark $\gamma > \frac{1}{2}$ would make them decay); spring 2 is assumed massless; the static calculation stops at the limit point (a restart of the Newton–Raphson on the fully damaged branch, or an arc-length method, would be needed to continue it quasi-statically).